

WIN11 OPTIMIZER

TESTING, QA & VERIFICATION GUIDE

Document 3 of 3 | QA Reference | v1.0

ASPECT	DESCRIPTION
Purpose	Verify each layer works correctly and system remains stable
Methodology	Pre-test snapshot → Apply → Verify → Observe → Confirm
Observation	48–72 hour observation window per layer (minimum)
Pass Criteria	All critical services running, no regressions, metrics improved
Fail Criteria	Any critical service broken, BSOD, update failure, boot failure
Recovery	Rollback script → System Restore → Disk Image

This document is the complete testing and verification framework for the Win11 Optimizer. Every layer has a dedicated test checklist, a set of PowerShell verification commands, and a pass/fail criteria table. Run these checks before proceeding to the next layer, and after any rollback to confirm the system has fully recovered.

01 | TESTING PHILOSOPHY

Testing a debloated system is not just about checking if Windows boots. It requires a structured verification approach: confirming protected services are intact, measuring actual resource improvements, and validating no hidden regressions appeared in subsystems you rely on.

TEST TYPE	DESCRIPTION	WHEN TO RUN
Integrity Check	Verify critical services still running	After every layer
Regression Check	Confirm no broken features you need	After every layer
Performance Check	Measure RAM/CPU/boot improvement	After every layer
Stability Check	Normal usage for 48-72 hours	Per layer, ongoing
Rollback Test	Verify rollback fully restores state	Before applying next
Security Check	Confirm Defender + Firewall active	After every layer
Update Check	Run Windows Update and confirm success	After Moderate+
Networking Check	Confirm Wi-Fi, Ethernet, DNS working	After firewall tweaks

02 | PRE-TEST BASELINE (Run BEFORE Any Layer)

These commands establish your system baseline. Run them before applying ANY layer and save the output. You'll compare against this baseline after each layer.

CRITICAL – Export all running services

```
Get-Service | Where-Object {$_.Status -eq 'Running'} | Select Name,DisplayName,StartType | Export-Csv C:\WinOptimizer\snapshots\baseline_running_services.csv -NoTypeInfo
```

Export all service startup types

```
Get-Service | Select Name,DisplayName,Status,StartType | Sort-Object Name | Export-Csv C:\WinOptimizer\snapshots\baseline_all_services.csv -NoTypeInfo
```

Export all provisioned AppX packages

```
Get-AppxProvisionedPackage -Online | Select DisplayName,PackageName,Version | Export-Csv C:\WinOptimizer\snapshots\baseline_appx.csv -NoTypeInfo
```

Export all scheduled task states

```
schtasks /query /fo CSV /v > C:\WinOptimizer\snapshots\baseline_tasks.csv
```

Export all firewall rules

```
Get-NetFirewallRule | Select Name,DisplayName,Direction,Action,Enabled | Export-Csv C:\WinOptimizer\snapshots\baseline_firewall.csv -NoTypeInfo
```

Record RAM baseline

```
$ram = Get-CimInstance Win32_OperatingSystem  
$usedMB = [math]::Round(($ram.TotalVisibleMemorySize - $ram.FreePhysicalMemory) / 1024)  
Write-Output "Baseline RAM Used: $usedMB MB" | Tee-Object C:\WinOptimizer\snapshots\baseline_ram.txt
```

Record process count baseline

```
Write-Output "Baseline Processes: $((Get-Process).Count)" | Tee-Object C:\WinOptimizer\snapshots\baseline_processes.txt
```

Record running services count

```
Write-Output "Baseline Running Services: $((Get-Service | Where {$_.Status -eq 'Running'}).Count)" | Tee-Object C:\WinOptimizer\snapshots\baseline_svc_count.txt
```

Create System Restore point

```
Enable-ComputerRestore -Drive "C:"  
Checkpoint-Computer -Description "pre-optimizer-baseline-$(Get-Date -f yyyyMMdd_HH:mm:ss)" -RestorePointType MODIFY_SETTINGS
```

Export current registry policies

```
reg export "HKLM\SOFTWARE\Policies" C:\WinOptimizer\snapshots\baseline_policies_HKLM.reg /y  
reg export "HKCU\SOFTWARE\Policies" C:\WinOptimizer\snapshots\baseline_policies_HKCU.reg /y
```

03 | POST-LAYER VERIFICATION SUITE

After applying any layer, run this verification suite. Every check must PASS before you proceed to the next layer. A single FAIL requires investigation and rollback before continuing.

CRITICAL SYSTEM INTEGRITY CHECKS (All Layers)

```
# CHECK 1 – Windows Defender must be ACTIVE
(Get-MpComputerStatus).AntivirusEnabled
# Expected: True

CRITERIA: PASS: True | FAIL: False → STOP, investigate immediately

# CHECK 2 – Windows Firewall must be ACTIVE
(Get-NetFirewallProfile -Profile Domain,Public,Private).Enabled
# Expected: True True True

CRITERIA: PASS: All True | FAIL: Any False → rollback firewall tweaks

# CHECK 3 – Windows Update service must be RUNNING
(Get-Service wuauserv).Status
# Expected: Running or Stopped (demand start)

CRITERIA: PASS: Running or Stopped | FAIL: Disabled → immediate rollback

# CHECK 4 – TrustedInstaller must be accessible
Get-Service TrustedInstaller | Select Name,StartType
# Expected: StartType = Manual

CRITERIA: PASS: Manual or Running | FAIL: Disabled → immediate rollback

# CHECK 5 – LSASS process must be running
Get-Process lsass
# Expected: Running process returned

CRITERIA: PASS: Process found | FAIL: System in broken state

# CHECK 6 – Run Windows Update – must succeed
# Manually: Settings → Windows Update → Check for Updates
# Or: Install-Module PSWindowsUpdate; Get-WindowsUpdate

CRITERIA: PASS: Updates download/install | FAIL: Error → investigate

# CHECK 7 – Network connectivity
Test-NetConnection google.com -Port 443
# Expected: TcpTestSucceeded = True

CRITERIA: PASS: True | FAIL: False → check firewall rules added

# CHECK 8 – Microsoft Store reachable (if not removed)
Test-NetConnection aka.ms -Port 443
# Expected: TcpTestSucceeded = True

CRITERIA: PASS: True | FAIL: False → firewall rule too aggressive
```

04 | PER-LAYER TEST CHECKLISTS

[LAYER 1 — MINIMAL] — POST-APPLY CHECKLIST

#	TEST ITEM	VERIFICATION METHOD	EXPECTED RESULT	PASS?
1	DiagTrack disabled	sc query DiagTrack	STATE: STOPPED	[]
2	Startup apps reduced	Task Manager → Startup tab	Fewer high-impact items	[]
3	Widgets absent from taskbar	Check taskbar	Widget icon gone	[]
4	No Start menu ads	Open Start menu	No 'Suggested' apps	[]
5	Lock screen = Picture not Spotlight	Personalization settings	Type = Picture	[]
6	Edge startup boost off	Edge → Settings → System	Startup boost = Off	[]
7	Delivery Opt = Off	Settings → WU → Delivery Optimization	P2P sharing off	[]
8	Defender still active	Get-MpComputerStatus	AntivirusEnabled = True	[]
9	Windows Update works	Check for updates	No errors	[]
10	RAM reduced	Task Manager → Performance	Lower than baseline	[]
11	Boot time improved	Startup time (Event Viewer)	Faster than baseline	[]
12	No BSOD in 48h usage	Check Event Viewer → Windows Logs → System	No critical errors	[]

[LAYER 2 — MODERATE] — POST-APPLY CHECKLIST

#	TEST ITEM	VERIFICATION METHOD	EXPECTED RESULT	PASS?
1	Xbox services disabled	Get-Service XblAuthManager	Status: Stopped, Disabled	[]
2	Telemetry policy set	Get-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\Windows\DataCollection"	AllowTelemetry=0	[]
3	CEIP tasks disabled	schtasks /query findstr Consolidator	Status: Disabled	[]
4	Xbox apps removed	Get-AppxPackage *Xbox*	No packages returned	[]
5	SysMain disabled (if SSD)	Get-Service SysMain	Status: Stopped	[]
6	Defender exclusions active	Get-MpPreference Select ExclusionPath	Folders listed	[]
7	OneDrive removed (if chosen)	Get-Process OneDrive	No process running	[]
8	WSearch still ok (moderate)	Get-Service WSearch	Running (not yet disabled)	[]
9	Windows Update succeeds	Run Windows Update	Updates download/install OK	[]
10	Network still functional	Test-NetConnection google.com -Port 443	TcpTest: True	[]
11	Office/productivity apps work	Open Word/Excel/browser	No errors, opens fine	[]
12	RAM reduced further	Task Manager	Lower than Minimal baseline	[]
13	No print issue (if used)	Try printing a test page	Prints OK	[]
14	48h stability check	Normal usage for 48 hours	No regressions	[]

[LAYER 3 — ULTIMATE] — POST-APPLY CHECKLIST

#	TEST ITEM	VERIFICATION METHOD	EXPECTED RESULT	PASS?
1	Firewall rules created	Get-NetFirewallRule Where DisplayName -Not \$null	Rules listed	[]
2	Blocked IPs not affecting Store	Open Microsoft Store	Store loads normally	[]
3	Provisioned apps removed	Get-AppxProvisionedPackage -Online	Target packages gone	[]
4	Hibernate off	powercfg /a	Hibernate not listed	[]
5	WerSvc disabled	Get-Service WerSvc	Status: Disabled	[]
6	Driver update GPO set	gpedit.msc or registry check	Policy value = 1	[]
7	System image created	Check external backup	Image file present	[]
8	Windows Update still works	Run Windows Update	Succeeds without errors	[]
9	All network adapters work	Device Manager → Network Adapters	No yellow warnings	[]
10	GPU drivers current	Device Manager → Display Adapters	No warnings	[]
11	Boot still successful	Reboot 3 times	All boots successful	[]
12	No app crashes (72h obs.)	Event Viewer → Application	No recurring errors	[]
13	Firewall not blocking Office	Open Outlook/Teams	Authentication works	[]
14	BitLocker unaffected (if used)	manage-bde -status	Protection On	[]

[LAYER 4 — GOD MODE] — POST-APPLY CHECKLIST

#	TEST ITEM	VERIFICATION METHOD	EXPECTED RESULT	PASS?
1	Consumer features blocked	Get-ItemProperty "HKLM:\SOFTWARE\Policies\Microsoft\WindowsFeatures" Where-Object FeatureName -Like "Content"	Disabling successful	[]
2	Copilot disabled	Check taskbar	Copilot button gone	[]
3	WSearch fully stopped	Get-Service WSearch	Status: Disabled	[]
4	Everything Search installed	Open Everything.exe	Instant search works	[]
5	OneSyncSvc disabled	Get-Service OneSyncSvc	Status: Disabled	[]
6	Print Spooler off (if chosen)	Get-Service Spooler	Status: Disabled	[]
7	Xbox fully removed	Get-AppxPackage *Xbox*; Get-Service Xbl* Where-Object Name -Not \$null	Empty + Disabled	[]
8	No background UWP tasks	Task Manager → Background Processes	Significantly reduced	[]
9	SysMain disabled	Get-Service SysMain	Status: Disabled	[]
10	RAM at target (~1.6-2 GB idle)	Task Manager	Close to target range	[]
11	CPU idle < 2%	Task Manager → Performance	Near 0-2% idle	[]
12	Background processes < 45	Get-Process Measure-Object	Count < 45	[]
13	Windows Update MUST work	Run Windows Update	Critical — must pass	[]
14	Defender active	Get-MpComputerStatus	AntivirusEnabled = True	[]
15	All drivers functional	Device Manager	Zero yellow warnings	[]
16	Boot time measured	Event Viewer → EventID 100	Record in ms	[]
17	72h stability observation	Normal + intensive usage	Zero critical errors	[]
18	Rollback tested (critical)	Run rollback_godmode.ps1	Reverts cleanly	[]

05 | AUTOMATED VERIFICATION SCRIPT

This script runs all critical integrity checks automatically and outputs a PASS/FAIL report. Run it after every layer application.

```
function Invoke-SystemVerification {
    param([string]$Layer = "All")
    $results = @()
    $allPassed = $true

    Write-Host "`n=== WIN11 OPTIMIZER - VERIFICATION SUITE ===" -ForegroundColor Magenta
    Write-Host "Layer: $Layer | Time: $(Get-Date -f 'yyyy-MM-dd HH:mm:ss')`n"

    # TEST 1: Windows Defender
    $defActive = (Get-MpComputerStatus -EA SilentlyContinue).AntivirusEnabled
    $r1 = [PSCustomObject]@{ Test="Defender Active"; Expected="True"; Actual=$defActive; Pass=($defActive -eq $true) }
    $results += $r1

    # TEST 2: Windows Firewall
    $fwProfiles = Get-NetFirewallProfile -Profile Public,Private -EA SilentlyContinue
    $fwActive = ($fwProfiles.Enabled -notcontains $false)
    $r2 = [PSCustomObject]@{ Test="Firewall Active"; Expected="All Enabled"; Actual=$fwActive; Pass=$fwActive }
    $results += $r2

    # TEST 3: Windows Update service
    $wuStatus = (Get-Service wuauserv -EA SilentlyContinue).StartType
    $r3 = [PSCustomObject]@{ Test="WU Service StartType"; Expected="Not Disabled"; Actual=$wuStatus; Pass=($wuStatus -ne "Disabled") }
    $results += $r3

    # TEST 4: TrustedInstaller
    $ti = (Get-Service TrustedInstaller -EA SilentlyContinue).StartType
    $r4 = [PSCustomObject]@{ Test="TrustedInstaller"; Expected="Manual or Auto"; Actual=$ti; Pass=($ti -ne "Disabled") }
    $results += $r4

    # TEST 5: Network connectivity
    $netTest = Test-NetConnection google.com -Port 443 -InformationLevel Quiet -EA SilentlyContinue
    $r5 = [PSCustomObject]@{ Test="Network (google:443)"; Expected="True"; Actual=$netTest; Pass=$netTest }
    $results += $r5

    # TEST 6: LSASS running
    $lsass = Get-Process lsass -EA SilentlyContinue
    $r6 = [PSCustomObject]@{ Test="LSASS Process"; Expected="Running"; Actual=($lsass -ne $null); Pass=($lsass -ne $null) }
    $results += $r6

    # TEST 7: RAM usage
    $ramObj = Get-CimInstance Win32_OperatingSystem
    $ramUsedMB = [math]::Round(($ramObj.TotalVisibleMemorySize - $ramObj.FreePhysicalMemory)/1024)
    $ramTotalMB = [math]::Round($ramObj.TotalVisibleMemorySize/1024)
    $r7 = [PSCustomObject]@{ Test="RAM Used"; Expected="< 75% total"; Actual="$ramUsedMB MB / $ramTotalMB MB"; Pass=($ramUsedMB -lt ($ramTotalMB * 0.75)) }
    $results += $r7

    # TEST 8: Running processes count
    $procCount = (Get-Process).Count
    $r8 = [PSCustomObject]@{ Test="Process Count"; Expected="< 120"; Actual=$procCount; Pass=($procCount -lt 120) }
    $results += $r8
}
```

```
# Print results
foreach ($r in $results) {
    $status = if ($r.Pass) { "[PASS]" } else { "[FAIL]" }
    $color = if ($r.Pass) { "Green" } else { "Red" }
    Write-Host "  $status $($r.Test.PadRight(28)) Actual: $($r.Actual)" -ForegroundColor $color
    if (-not $r.Pass) { $allPassed = $false }
}

Write-Host ""
if ($allPassed) {
    Write-Host "  ALL CHECKS PASSED - System integrity confirmed." -ForegroundColor Green
} else {
    Write-Host "  FAILURES DETECTED - Do NOT proceed to next layer." -ForegroundColor Red
    Write-Host "  Review failures above and run rollback if needed." -ForegroundColor Yellow
}

# Export to file
$results | Export-Csv "C:\WinOptimizer\logs\verify_$(Get-Date -f yyyyMMdd_HH:mm).csv" -NoTypeInformation
return $allPassed
}

# Run it:
Invoke-SystemVerification -Layer "Moderate"
```


06 | REGRESSION TESTS — BY FEATURE

These feature-specific regression tests confirm you haven't broken anything you actually use. Run the tests relevant to your workflows after each layer.

FEATURE AREA	TEST	HOW TO TEST	AFFECTED LAYER
Windows Update	Updates download + install	Settings → WU → Check	All layers
Wi-Fi / Network	Connect to Wi-Fi, browse web	Open browser, ping	Ultimate+
Microsoft Store	Open and browse Store	Open Store app	Ultimate+
Office Suite	Open Word/Excel, save, auto-save	Create + save document	Moderate+
Microsoft Account	Sign in/out of MS account	Settings → Accounts	God Mode
Printing	Print a test page	Control Panel → Devices	God Mode
Bluetooth	Pair and connect a device	Settings → Bluetooth	Moderate+
USB Devices	Plug in USB drive, access files	File Explorer	All layers
Audio	Play audio, system sounds	VLC / YouTube	All layers
Display	Adjust brightness, resolution	Settings → Display	Ultimate+
Sleep/Wake	Sleep and wake laptop	Close lid, reopen	Ultimate+
Fast Boot	Reboot and time it	Compare to baseline	All layers
Driver Updates	Check Device Manager	No yellow warnings	Ultimate+
Gaming (if used)	Launch a game	Run a game for 10 min	God Mode
Dev Tools	Run Python/Node/Docker	Run a test build	Moderate+
File Indexing	Search for a file by name	Start menu search	God Mode
Notifications	App notifications appear	Use any app briefly	Minimal+
Virtual Memory	Run a heavy workload	No out-of-memory errors	Ultimate+

07 | PERFORMANCE MEASUREMENT COMMANDS

Use these commands to quantitatively measure improvement after each layer. Record the values in the comparison table below.

```
# Measure RAM used (MB)
$R = Get-CimInstance Win32_OperatingSystem
[Math]::Round(($R.TotalVisibleMemorySize - $R.FreePhysicalMemory) / 1024)

# Count all running processes
(Get-Process).Count

# Count all running services
(Get-Service | Where-Object {$_.Status -eq 'Running'}).Count

# Get CPU idle % (quick sample)
(Get-WmiObject Win32_Processor | Measure-Object -Property LoadPercentage -Average).Average

# Get last boot time + uptime
$boot = (gcim Win32_OperatingSystem).LastBootUpTime
$uptime = (Get-Date) - $boot
"Boot: $boot | Uptime: $($uptime.ToString('hh:mm:ss'))"

# Count background (non-user) processes
Get-Process | Where-Object {$_.MainWindowHandle -eq 0} | Measure-Object | Select Count

# Check disk I/O (brief sample)
Get-Counter '\PhysicalDisk(_Total)\Disk Bytes/sec' -SampleInterval 2 -MaxSamples 5 | Select -ExpandProperty CounterSamples | Select CookedValue

# Count disabled services (verify tweaks applied)
(Get-Service | Where-Object {$_.StartType -eq 'Disabled'}).Count

# Export full current service state (for comparison)
Get-Service | Select Name,Status,StartType | Export-Csv C:\WinOptimizer\logs\services_post_$(Get-Date -f yyyyMMdd_HHmm).csv -NoTypeInformation
```

PERFORMANCE TRACKING TABLE (Fill In Manually)

Record your measurements here after each layer:

Metric	Baseline	After Min.	After Mod.	After Ult.	After God
RAM Used (MB)	_____	_____	_____	_____	_____
Process Count	_____	_____	_____	_____	_____
Running Services	_____	_____	_____	_____	_____
CPU Idle %	_____	_____	_____	_____	_____
Disabled Services	_____	_____	_____	_____	_____
Boot Time (sec)	_____	_____	_____	_____	_____
Disk I/O Idle (KB)	_____	_____	_____	_____	_____

08 | ROLLBACK VERIFICATION TESTS

After running a rollback, you must verify the system has fully recovered. A rollback is only successful when ALL these checks pass.

CHECK	COMMAND / METHOD	EXPECTED
Services restored	Compare services CSV to baseline	Matching startup types
AppX packages restored	Get-AppxPackage vs baseline CSV	Removed packages back
Registry values removed	Check specific keys from rollback script	Values deleted/restored
Scheduled tasks re-enabled	schtasks /query findstr TaskName	Status: Ready
Firewall rules removed	Get-NetFirewallRule Where DisplayName -match 'Block Block Telemetry'	Block Block Telemetry' rules
Defender still active	Get-MpComputerStatus	AntivirusEnabled: True
Windows Update works	Run Check for Updates	No errors
Network connectivity	Test-NetConnection google.com -Port 443	TcpTestSucceeded: True
Performance normal	Check RAM/CPU vs baseline	Similar to baseline
Boot successful x3	Reboot 3 times	All boots succeed
System Restore available	rstrui.exe → check restore points	Points listed

09 | KNOWN ISSUES & QUICK FIXES

Common post-tweak issues and their immediate fixes:

SYMPTOM	LIKELY CAUSE	QUICK FIX
Windows Update fails (0x80070005)	Delivery Optimization disabled	Re-enable via Settings or registry
Microsoft Store won't open	Store framework removed	Re-register: Get-AppxPackage *Store* Add-AppxPackage
No Start menu search results	WSearch disabled	sc start WSearch && sc config WSearch start=delayed-auto
Wi-Fi disconnects randomly	Firewall rule too broad	Remove-NetFirewallRule matching broadest rule
Office activation fails	Firewall blocking licensing	Check/remove firewall rules blocking port 443
Xbox Game Pass login fails	Xbox services disabled	Set-Service XblAuthManager -StartupType Manual; Start-Service XblAuthM
System won't sleep/hibernate	Hibernate disabled	powercfg -h on
Print dialog crashes	Spooler disabled	sc config Spooler start=auto && sc start Spooler
Bluetooth stopped working	BTH service affected	Check Get-Service bthserv — enable if disabled
Cortana taskbar search broken	Cortana disabled but Search needs it	Separate WSearch from Cortana if possible
Settings sync lost	OneSyncSvc disabled	Set-Service OneSyncSvc -StartupType Manual; Start-Service OneSyncSvc
Can't install new apps (Store)	Provisioned packages corrupted	In-place repair install from Windows 11 ISO
Game crashes immediately	Xbox overlay removed	Reinstall Xbox Game Bar from Store

RAM higher than expected	Some services not disabled	Re-run verification, check sc query for each service
--------------------------	----------------------------	--

10 | FINAL GO / NO-GO CRITERIA

Before declaring any layer 'complete' and moving to the next, this Go/No-Go gate must be passed. Every single GO item must be confirmed.

CRITERIA	STATUS	GATE
Windows Defender is ACTIVE and scanning	☐ Y/N	GO
Windows Firewall is ENABLED on all profiles	☐ Y/N	GO
Windows Update ran successfully (no errors)	☐ Y/N	GO
Network connectivity confirmed (DNS + HTTPS)	☐ Y/N	GO
No BSOD or critical errors in Event Viewer	☐ Y/N	GO
48-72 hour observation period completed	☐ Y/N	GO
System boots cleanly x3	☐ Y/N	GO
All drivers in Device Manager show no warnings	☐ Y/N	GO
Rollback script tested and confirmed working	☐ Y/N	GO
System Restore point exists for this layer	☐ Y/N	GO
RAM usage improved vs baseline	☐ Y/N	BONUS
CPU idle usage improved vs baseline	☐ Y/N	BONUS
No regressions in workflows you actually use	☐ Y/N	GO

DECISION MATRIX

- ALL [GO] items = YES
- PROCEED to next layer
- ANY [GO] item = NO
- STOP. Investigate + rollback if needed.
- [BONUS] items = optional
- Record value, no blocking requirement
- Rollback not tested
- MANDATORY before God Mode

This completes the 3-document Win11 Optimizer production reference. Document 1: System Architecture | Document 2: Implementation Strategy | Document 3: Testing & Verification. Together these form the complete engineering specification for building and running the tool safely.